AS notes

Q) Why are B^- particle emitted with range of Kinetic energies
- anti-neutrinos are emitted so energy is shared.

Q) Two conditions for an object to be in equilibrium
- resultant force is zero.
- resultant torque is zero.

Extension and wavelength have linear relationship.

Potential difference – Energy transferred from electrical to other forms per unit charge.

The closer the block to the beam, the lesser tension in the wire.

Stationary waves should travel in opposite directions and overlap each other, have same frequency.

Despite the masses, if air resistance is negligible, speed and acceleration remains the same.

- Principle of conservation of momentum – momentum before collision = momentum after collision in a closed system.
- Coherent – Constant phase difference.

⇄ Shorter wavelength, smaller slit separation.

- Show $R = R_1 + R_2 + R_3$

In series, current stays the same.

$V = IR \quad V = V_1 + V_2 + V_3$

$\cancel{I}R = \cancel{I}R_1 + \cancel{I}R_2 + \cancel{I}R_3$

$R = R_1 + R_2 + R_3$

Q) Explain why horizontal component of velocity remains constant?

force in horizontal direction is zero.

When no resistive force, KE = GPE

Work done = GPE − KE (changes in energ

$E = \frac{1}{2}K(x^2)$.

↳ square shows quadratic, so graph is a curve.

- Phase difference between 2 adjacent nodes = 0°
- Power – work done / time taken

- ΔKinetic energy = 0 when velocity is constant

Work done is the total energy required to move an object.

- Speed of sound in air = $338 ms^{-1}$
- A stationary wave can be detected when a louder sound is heard.
- number density of free electrons = $\frac{\text{no. of electrons}}{\text{volume}}$

$\text{resistance} \propto \frac{1}{\text{drift speed}}$

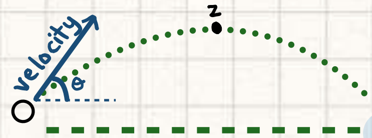
$$Km \xrightarrow{\times 1000} m \xrightarrow{\times 100} cm \xrightarrow{\times 10} mm$$

$$mm \xrightarrow{\div 10} cm \xrightarrow{\div 100} m \xrightarrow{\div 1000} Km$$

$(\_)^2 = (100)^2$

$(\_)^3 = (100)^3$

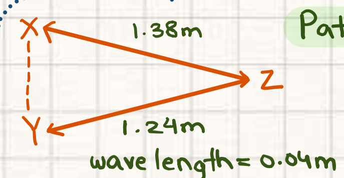
- Draw out the direction of forces
- upthrust increases so stress increases. hence strain increases.



speed at z

Speed = horizontal component of velocity = initial speed $\cos\theta$

Principle of superposition – when waves overlap, the resultant displacement is the sum of individual displacements.



Path diff = $\frac{XZ - YZ}{\lambda}$

$= \frac{1.38 - 1.24}{0.04\lambda} = 3.5\lambda$

If path difference is in whole number, constructive interference phase diff = 0°, 360°

If path difference is in $(n + \frac{1}{2})$, destructive interference phase diff = 180°

At destructive interference, intensity is zero.

Define ohm? volt / ampere

The gradient of a $\frac{V}{I}$ graph shows internal resistance

Q) Current = 6.9×10^{-9} A   Time = 1 min
find number of α-particles passing.

22/m/j/2021
qu-6

$Q = ne$
$It = ne$
$$n = \frac{(6.9\times10^{-9})(60)}{2(1.6\times10^{-19})} = 1.3\times10^{12}$$
alpha particles have charge of +2.

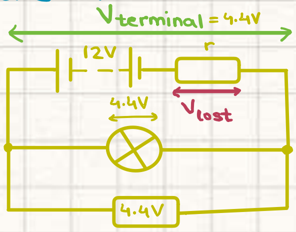
Distance is area under velocity-time graph, acceleration is just $\frac{v-u}{t}$.

Force - rate of change of momentum.

Before a object reaches constant terminal velocity, acceleration decreases as air resistance and weight increases, resultant force decreases

↑ density ∝ ↑ upthrust
↑ density ∝ ↓ extension

For questions with circular path, distance is $2\pi r$.



$E - V_{lost} = V_{terminal}$ (12V) (4.4V)

Drag / Viscous force is always opposite to the direction of motion/weight.
↳ drag force increases with speed so acceleration decreases.

Acceleration and time are inversely proportional.

If amplitude is doubled, intensity increases by a factor of 4.

Wavelength - Distance between 2 adjacent wavefronts

Newton's third law - Force applied by body B on body A is equal in magnitude and opposite in direction to body B.

In a perfectly elastic collision, the ratio is 1.

t = Time period x Wavelength.
or time-base setting → $T \times \lambda$

$\frac{\text{Path difference}}{\text{wavelength}}$ = Path difference in terms of λ.

Acceleration - change in velocity/time.

emf - energy transferred from chemical to electrical

P.d - energy transferred from electrical to thermal

$Q = ne$

learn the masses of particles.

frictional force acts between two surfaces.

$I = anve$
↳ n = number of free electrons per unit volume.

Antinode - maximum amplitude

Kirchoff's first law is linked to charge.

$E = IVt$

During nuclear decay, Kinetic energy is mostly transferred.

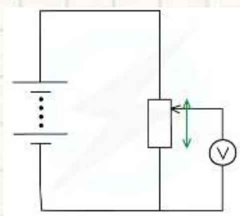
K.E - energy of a body due to speed.

constant velocity and force

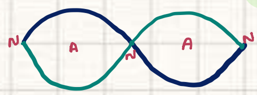
Driving force = Resistive force + weight

Doppler effect - Change in frequency when source moves relative to observer.

Go through 9702/23/O/N/22



If the slider is moved upwards, the resistance of the lower part will increase so potential difference will also increase.



All the points between two nodes have phase difference of 0 in stationary waves.

Interference – sum of displacements of overlapping waves.

If we don't know direction of mass after collision, consider it positive.

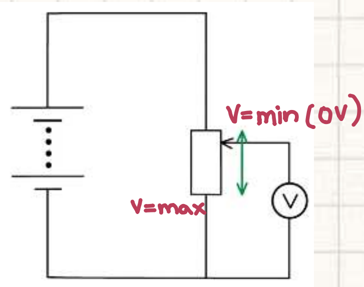
↳ after finding out velocity, if ans is

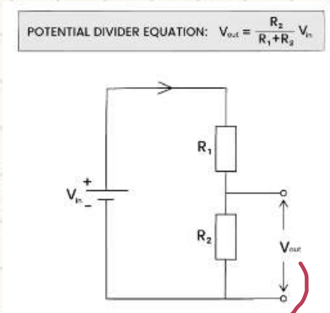
+ right

− left

• Potential difference ∝ Resistance

Potential divider only series circuit





To measure V across R_2.

Kirchoff's 1st law – charge

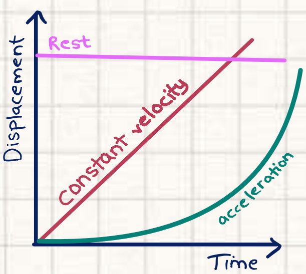
2nd law – conservation of energy.

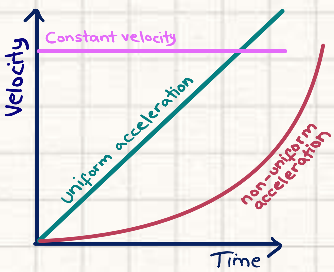
$$\text{Intensity} = \frac{\text{Power}}{\text{area}}$$

The results of the α-particle scattering experiment led to the development of the nuclear model for the atom.

State the results that suggested that most of the mass of the atom is concentrated in a very small region and most of the atom is empty space.

A small proportion of alpha particles are deflected back by angles greater than 90° and a large proportion of particles pass straight through





• Gradient – Velocity

• Gradient – Acceleration

• Area under graph – Displacement

Diffraction grating:

$n = \frac{d}{\lambda}$ — highest order of maxima

$d = \frac{1}{N}$ — number of lines on grating

$n\lambda = d\sin\theta$

Electricity:

$I = Anvq$ $q = 1.6\times10^{-19}C$

$V = \frac{W}{Q}$ $P = IV$ $P = I^2R$

$P = \frac{V^2}{R}$ $V = IR$ $R = \frac{\rho L}{A}$

$E = I(R+r)$

$V_{out} = \frac{R_2}{R_1+R_2} V_{in}$ — V we are measuring

When the object is moving towards the observer, frequency is higher.

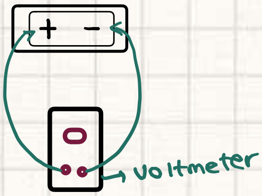
Made by: Ayesha Hayat

Q1)

- whenever finding time, take multiple readings.
- 5 oscillations atleast.
- if they say increase ___ and repeat experiment, take values after minimum.
- Check crocidile clips and battery.

↳ for example, in the qu. if they say battery is of 1.5V, attach wires to battery and voltmeter to check.

Check for all crocodile clips



- In table, write SI units.
- Keep significant number same.
  - ↳ decimal places
  - ↳ add zero if decimal place is less.
- Best-fit line - same number of points below and above line.

↳ far point - circle it and write anamalous point

- Graph - gradient tells units.

Time measurement - 1 decimal

Absolute uncertainty gets multiplied and divided

↳ fractional and percentage doesn't.

Write answers with precision.

↳ Length - 30.0 cm

6 sets of reading.

Table for oscillation

S.no	n	t_1/s	t_2/s	t_{avg}/s	T/s

1)e) equation uses $y=mx+c$ and include units.

Q2)

- for the first part, write absolute uncertainty with answers.
- Justify significant figures.

As least significant number in our raw value of length/etc are ___ significant figures.

$$\frac{K_1 - K_2}{K_{avg}} \times 100$$

↳ if difference <20%, relationship is valid.

Error:

↳ Two readings are not enough to draw a conclusion.

Improvement:

↳ Take multiple readings and plot a graph.

To find absolute uncertainty: (repeated)

- Take two readings
- Subtract them and divide by 2

for example: $h_1 = 20.7$ cm, $h_2 = 21.5$ cm | $h_{avg} = \frac{20.7+21.5}{2} = 21.1$ | $21.5 - 20.7 = 0.8$, $\frac{0.8}{2} = 0.4$

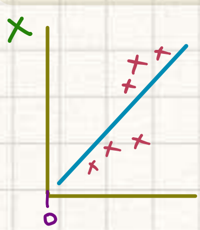
$h = 21.1 \pm 0.4$ cm

$$\% \text{ uncer} = \frac{\text{absolute}}{\text{value}} \times 100$$

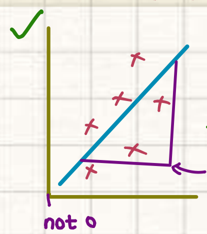
Try to take repeated readings.

- For table in raw values (like t_1, t_2), number of decimal places should be kept same. (like 9.1, 9.8, 10.1, 12.6)
- For calculated table values (t_{avg}), significant figures should be same, decimal places don't really matter.

Points should be crosses



(y-intercept can be directly read).



(y-intercept has to be calculated using $y=mx+c$).

Points shouldn't be in a order, like first 3 down and last 3 above.

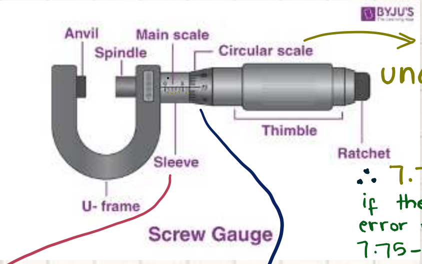
Triangle should be big

... Qu. 2

↳ 20V multimeter

↳ if the meter reading is negative, switch the wires.

↳ positive end of the battery should be connected to the positive terminal.



Tells uncertainty

∴ 7.75mm

if there's a zero error of 0.02, then $7.75 - 0.02 = 7.73$ mm

Tells the main reading

if scale is above zero, positive error.

Angles can be written as $14' \pm 2'$

Oscillations

↳ Error

- Difficult to judge exactly when an oscillation is complete

↳ Improvement

- Use a fiducial pointer.